$$AC = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} Ac_1 & Ac_2 \end{bmatrix}$$ (By defn. of matrix multiplication)

$$Ac_1 = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 5 \\ 3 \end{bmatrix} = 5\begin{bmatrix} 2 \\ -4 \end{bmatrix} + 3\begin{bmatrix} -3 \\ 6 \end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix} 10 \\ -20 \end{bmatrix} + \begin{bmatrix} -9 \\ 18 \end{bmatrix}$$ (Defn. of scalar multiple)

$$= \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$ (Vector Sum)

$$Ac_2 = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\cdot(-2)+(-3)\cdot 1 \\ (-4)\cdot(-2)+6\cdot 1 \end{bmatrix} = \begin{bmatrix} -7 \\ 14 \end{bmatrix}$$

(Row-Vector Rule)

$$\therefore AC = \begin{bmatrix} 1 & -7 \\ -2 & 14 \end{bmatrix}.$$ $\boxed{\therefore AB = AC}$ $\square$

11. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$. Compute AD and DA. Explain how the columns or rows of A change when A is multiplied by D on the right or on the left. Find a 3×3 matrix B, not the identity matrix or the zero matrix, such that $AB = BA$.

$\boxed{\text{Solution:}}$ $AD = \begin{bmatrix} Ad_1 & Ad_2 & Ad_3 \end{bmatrix}$ (By defn. of matrix multiplication)

$$Ad_1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 0\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + 0\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$ (By defn. of scalar multiple) $$= \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$ (By defn. of vector sum)

The 1st column of AD is a linear combination of the columns of A where the weights are the corresponding entries of the 1st Column of D.